Making Desk Compartments

Techniques for cubbies, drawers, and secret boxes

TEXT AND PHOTOS BY DOUG STOWE

This particular desk was made for an artist friend, Eleanor Lux, when her father's roll top desk, which she had used for many years, was inherited by her nephew. There were a number of considerations in designing the desk. Eleanor was tired of the roll top in her earlier desk and wanted something to open more easily, so I designed a paneled lift lid which utilized sliding door hardware and brass butt hinges. She wanted the lower drawers on both sides to hold standard files, allowing the desk to take the place of filing cabinets. Pull out "biscuit boards" were requested to provide additional working space during tax season and to facilitate organizing business mailings. No lap drawer is complete without a place for pencils. I make pencil trays using a simple cove cut technique on the tablesaw.

Eleanor had requested secret compartments and had originally wanted a space for a computer built into the design. I was relieved when she bought a new lap top computer and I no longer had to plan something that would become obsolete. In designing the desk cubbies, I was largely left to my own discretion. I scaled the compartments to fit standard sized papers and envelopes. Since I seldom make the exact same design of furniture twice, making cubbies that can be afforded by customers in my small town demands a simple approach. I have a straightforward technique for making cubbies using the router to cut the blind dadoes and tenons for the parts to fit together.



"Eleanor's Desk," made of solid cherry and maple, with Baltic birch ply drawer bottoms and back, would be too massive for most rooms, but fits the character of Eleanor's grocery-store-turned-weaving-studio, providing a special place for business organization within an artful environment.

One of the design principles I follow in my work is that of reflecting common elements in a variety of scales. I used large turned quarter columns on the outside of the desk to hide compartments for drawing tablets, poster board and the like. I used similar, but much smaller quarter columns to hide the secret compartments inside the desk. This maintains a continuity from first view to a more intimate relationship with the piece. While the large quarter columns are hinged to open on mini European-style hinges, the small quarter columns are affixed to secret boxes that slide out only after drawers are removed. As a serious box-maker, I couldn't resist making secret boxes part of the plan. To illuminate the desktop, I used

recessed quartz halogen low voltage fixtures built into the top of the cubby unit.

While the desk itself is far more complicated than could be described in an article, the technique used in making the simple cubbies may be of use to others.

I begin by gluing up the larger panels for the top and base from narrower stock. To help in the alignment of the boards, I use biscuits. I alternate the grain orientation to minimize the impact of changes in humidity. To prepare the stock for routing and cutting to shape, I use a drum sander, sanding to 150-grit. The drum sander doesn't give an ideal surface, as it still requires a great deal of hand and orbital sanding, but it is capable of attaining a uniform thickness without the tear-out that can come from planing wood with grain reversal.

In order to rout the top and bottom of the unit accurately, I clamp both sections together for the layout and routing. I use a tape measure and marking gauge to lay out the start and stop points for the router, and with a guide piece clamped firmly in place, I cut the mortises for the vertical elements to fit with a plunge router (**1**). To cut the bottom piece to shape, I use a series of templates clamped in place and a pattern bit in a laminate trimmer (**2**). The compact size of the laminate trimmer allows for getting in closer to the clamps holding the template pieces in place. I have to move the clamps to new positions out of the way as the cut proceeds. When the first shallow cuts are complete, the template can be removed to finish the cut, as the cut edge gives support to the pilot bearing. Prior to routing, I cut away most of the waste using the bandsaw, allowing the laminate trimmer and router bit to remove only a small amount of material at a time. I used a 1/2" radius roundover bit to shape the edge of the base, and then used a flat chisel to cut into the areas where the roundover bit could not reach (**3**), and finish the detail with a flat rasp and sandpaper.

A crosscut sled on the tablesaw with a stop block clamped in place insures that certain parts are cut to a uniform length. To rout the vertical parts for shelves and dividers to fit, I use a router table and straight-cut 1/4" router bit, with stop blocks clamped in place to control the movement of the stock across the table while the fence controls the position of the cut (**4**). To cut for additional shelves, the fence must be moved and the stop blocks must be relocated. I choose to do this with the router table rather than with a dado on the tablesaw because it allows the cut to be blind on the front edge. To rout the ends to fit the mortises, I use the router table with a larger straight-cut router bit. Using a tall fence, I pass the stock between the fence and cutter. A large cut made like this could present danger, pulling the stock and fingers into the cut, but because I am shearing away a very small amount I consider this a safe task. Nevertheless, I use safety blocking clamped to the router table, covering the cutter and protecting the fingers. Here, the safety blocking is removed to better show the process (**5**). The panels are then notched on the tablesaw to fit the stopped grooves (**6**).

Before final assembly, sanding, and finishing, I use a plunge router to cut finger holes to allow access for opening the secret









Cubbies are made from solid cherry and sized to hold standard paper and envelopes.

"Biscuit boards" slide out for additional working space. They are frame-and-panel construction to prevent warping, and slide on hardwood runners.

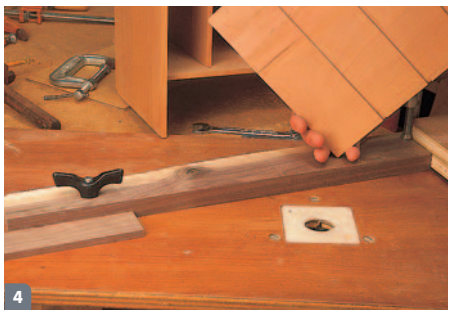
The maple pencil tray is shaped by using an angled tablesaw fence to create a cove cut. The dividers are fitted into slots cut with the plunge router.

Simple hinged boxes slide out after the drawers are removed. The curved moldings reflect the larger quarter columns on the outside of the desk.

















1. I use a marking gauge to lay out the start and stop points for routing the top and bottom for the cubbies to fit. The top and bottom are clamped together so they will line up at assembly time. I rout, pulling the router toward me so the rotation of the cutter keeps the router tight to the fence.
2. Templates clamped in place guide the laminate trimmer and pattern bit in shaping the base for the cubbies. When the first cut is made deep enough, the template can be removed and the first cuts will guide the subsequent cuts to the full depth of the stock. To avoid tear-out, I make the cuts in shallow increments and trim away most of the excess with the bandsaw.
3. Use a straight chisel to clean up the small radius left by the router. I approximate the contour of the roundover bit by making a series of straight cuts. A rasp and sandpaper are useful for cleaning up the chisel cuts and giving smooth definition to the shape of the base.
4. Use a straight-cut bit in the router table to cut the cubby sides for the shelves and dividers to fit. The stop blocks and fence provide the means to control the length and position of the cut. Each change of position in the fence requires a change in the stop blocks, and the stop blocks must be changed to rout opposite sides.
5. Holding the workpiece vertical between the fence and cutter, form tenoned ends on the shelves, dividers, and vertical parts for the cubbies to fit the routed dadoes. This is a climb cut, meaning that the cutter is pulling into the stock. This procedure can be dangerous, pulling the fingers in as well, but in this case it is reasonably safe because of the small cut. I use safety blocking to protect my fingers from accident. Here the safety blocking is removed to show the nature of the cut.
6. To allow the shelves and dividers to fit the blind dadoes, I make additional cuts on the tablesaw sled. The stop block is carefully set up so the cut aligns with the edge of the cut tenon.

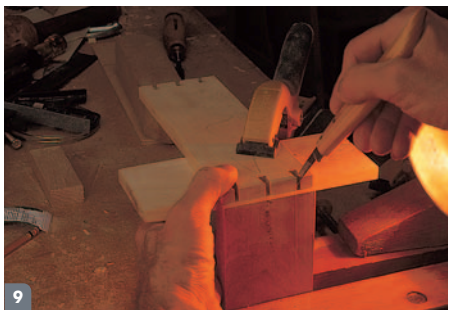
compartments. This is done with the plunge router, 1" straight-cut bit, the fence to control the placement of the cut, and penciled start and stop lines.

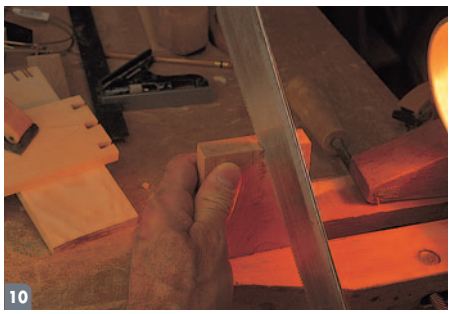
In complicated assemblies such as these cubbies, I find it helpful to do as much finish work as possible prior to final assembly. An accordion-style squeeze bottle reduces mess during glue-up (**7**) as I clamp the cubbies together in sections (**8**).

Making the drawers after the main assembly is complete allows me to more accurately measure their fit. I start by cutting the tails in the sides, and then with a block clamped in place to help me position it on the drawer front, I use a knife to mark for cutting the pins (**9**). I use a *dozuki* saw to make the angle cuts for the half-blind dovetails, taking care to only cut between the marks left by the marking gauge (**10**). As described in an earlier article in *Woodwork* [**August 2001**], I use a plunge router to first remove the waste between the hand-sawn cuts (**11**), followed by a straight chisel to finish forming the pins in the cherry drawer fronts. I make a groove for the drawer bottoms to fit the sides, front and back on the router table. Using the router table allows stop blocks to be clamped in place to control the length of cut so the groove does not appear on the outside of the drawers (**12**). I use 1/4" Baltic birch plywood for the drawer bottoms and apply glue carefully to the pins and tails as the drawers are assembled (**13**).

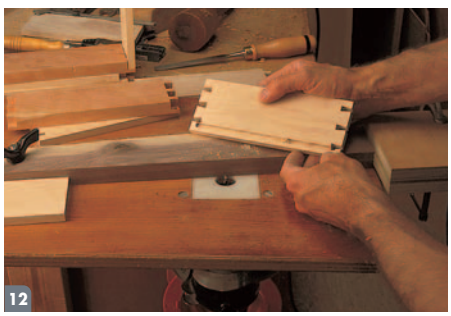
To begin making the drawer pulls, I use a laminate trimmer and template-following dado clean-out bit to rout the recess behind the pulls. The first cuts are made with the template in place, with the subsequent cuts made using the first cuts to provide bearing surface for the bit (**14**). This is the same process used to rout the shape of the base. I use the router table to begin shaping the drawer pulls. A 1" straight-cut bit protruding slightly from the fence is used to create the radiused cut as the workpiece is moved between stop

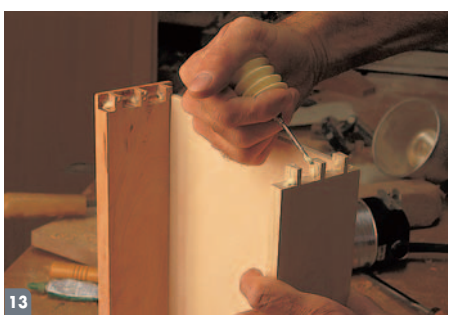
7. An accordion squeeze bottle is useful for applying glue to the insides of the stopped dadoes. Use plenty of clamps, cushioned with wooden blocks, to hold the units together as the glue sets.
8. It helps to do as much finish work as possible prior to assembly. The assembled, sanded, and finished units slip into the blind dadoes in the base.
9. I use a knife to transfer the shape of the tails onto the drawer fronts prior to cutting the pins. The maple block clamped to the drawer side is aligned with the bottom of the dovetails and helps to position the piece for accurate marking.
10. I use a *dozuki* saw to cut the pins. The cut is made at an angle and restricted to the space between the marks made by the marking gauge.
11. I use a plunge router to begin removing the waste from between the cuts. The routed area leaves a flat staging area for the finish chiseling and fitting of the joint.
12. Use of the router table to cut blind grooves for the drawer bottoms also speeds up work. Stop blocks control the length of the cut.
13. I spread glue liberally on the pins and tails prior to assembly.
14. A simple template clamped in place makes routing for the pulls easy. I use a "dado clean-out" bit in the laminate trimmer. To make the template, I used a 3/4" straight-cut router bit in a plunge router with fence. To finish the cut to depth, remove the template and use the earlier cut to guide subsequent cuts.
15. Shaping the pulls is done on the router table using a 1" straight-cut router bit almost completely buried in the fence. Stop blocks control the length and position of the cut. A 1/8" radius roundover bit buried in the fence is used for shaping the pulls.
16. Stop blocks control the length of cut for cutting attachment slots in the backs of the pulls. I use a 3/16" straight-cut router bit.


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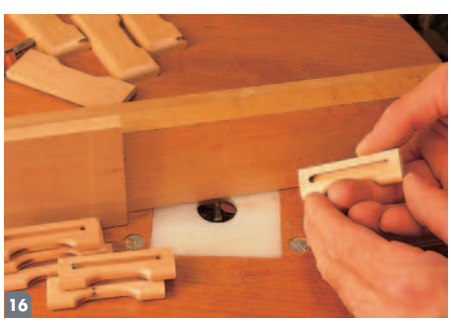

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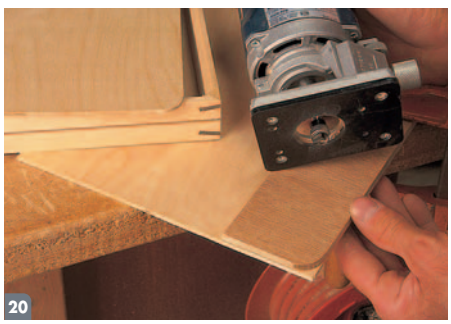
blocks. To keep the pulls from splitting, the cut is made only part way, then the piece is flipped over to complete the cut (**15**). Additional shaping is done with a roundover bit in the router table. Working close to the bit can be dangerous and I keep the bit buried inside the router fence as much as possible to minimize possible exposure to the bit were my fingers to slip. A blind groove at the back of the pull allows it to be attached to the drawer with a wood strip between. To cut the dado, I use the router table and move the workpiece between stops. The 3/16" straight-cut router bit is raised about 1/8" above the table (**16**). I sand the pulls using an inverted orbital sander. In this case, the sander is turned off, with the pad cushioning the work, but with the movement of the fingers supplying the energy. For very small parts, the motion of the sander is less effective than human power (**17**). To attach the pulls, I rout a matching groove in the drawer front and cut a strip to connect the pull to the drawer (**18**). With a bit of glue in the grooves, and the strip in place, I clamp the pulls to the drawer front. I check carefully for any glue that squeezes out, but wait for it to become leather hard before removing it with a chisel.

Making the secret boxes starts with assembling the box with mitered corners. After the corners are glued, I cut them for walnut splines or "slip feathers" to fit. I like to cut them at a slight angle to add interest, and because it gives slightly greater surface area to the joint. Two of these "slip feathers" at opposing angles give an amazing strength to a joint. Using a combination blade on the tablesaw gives a square crown to the cut. My jig for use on the tablesaw adjusts for the spacing of the cuts by moving a stop block, so every cut passes through the jig at the same point regardless of changes in the position of the stock. This keeps the cut clean and crisp (**19**). I glue small triangles of walnut in place to secure the miter

















17. Hand sanding on a turned-off inverted orbital sander. The pad from the sander helps. If the sander was turned on, however, it would make the parts difficult to handle and tiring to control.

18. Use a small block of wood to connect the pull to the drawer. The groove in the drawer was cut using the plunge router. Wood pads keep the clamp from leaving marks.

19. Use a slip feather jig on the tablesaw to cut for walnut splines to fit the mitered corners of the "secret" boxes. The jig uses the tablesaw guide slots and maple runners. The cut is always made into the same spot on the jig body, making certain that the cut is supported against unnecessary tear-out. Adjustment of the position of the cut is made with a moveable stop block. I squeeze glue into the saw cuts and "slip" the feathers in place. Next, they will be sanded flush on the stationary belt sander.

20. I use a plywood template and a pattern bit to shape the top and bottom panels to fit the rabbeted box sides. Note the saw cuts made in the box sides, just deep enough that a razor knife can finish the job.

21. Use the router table to cut the box base and cherry molding to fit the plywood bottom. I use a 3/4" straight-cut router bit, fence and stop blocks to control the position and length of cut. The straight-cut bit is raised above the table to match the thickness of the 1/8" Baltic birch plywood.

22. I use a brad nailer and glue to attach the plywood panels. The nail holes will be filled prior to finishing. Note how the box bottom panel extends for attaching the cherry molding.

23. I use a plunge router to rout a finger pull for removing the boxes from their places between the cubbies. The cut is just deep enough to catch with the finger tip. To find the location for the cut, I simply placed the box in position and traced the opening in the housing with a pencil, then routed between the penciled lines.

24. The slot hinges press in place to hold the box lid to the base. Butt hinges would work as well.

joints. I use a rabbeting cutter in the router table to cut the box sides for the top and bottom to fit. To shape the top and bottom panels to fit, I use a pattern bit in the router and a guide piece shaped from scrap plywood. After fitting the guide piece to a corner of the box, it can be used to exactly duplicate the same corners on all the required parts (**20**). Note the line sawn in the box for the top and bottom sections to be separated later. Just before assembly, I separate the top and bottom of the box using a razor knife to finish the cut.

Fitting the cherry molding to the box bottom requires some additional steps done on the router table with a straight-cut router bit. The fence and stop blocks control the position and length of cut (**21**). These cuts allow the box bottom to connect the frame of the box to the molding strips. I shaped the molding strips from solid cherry stock using a 1/2" round-over bit in the router table. I use a brad nailer and glue to attach the box top and bottom to the frame. Note in the photo how the box bottom extends beyond the frame to allow for it to be glued and nailed to the cherry molding (**22**). I use a plunge router to rout a small finger pull in the box (**23**).

As in many of my boxes, I use "barbed" or "slot" hinges to connect the box tops and bottoms. These hinges have teeth and slip into 3/64" slots cut into the box base and lid, and require a chamfer at the back side to allow clearance for opening (**24**). [**See *Woodwork* #66, pp. 62-66**] Butt hinges would work as well for this job.

Eleanor's desk is finished with Deft Danish Oil. I pass by Lux Weaving Studio on my evening walks and it always gives me pleasure to see it in use.

Doug Stowe is a contributing editor for Woodwork *magazine.*